

Restaurant Web Application Project **Outline**

1. Introduction

This document outlines the project plan for developing a comprehensive Restaurant Web Application. The primary goal of this project is to create a modern, user-friendly platform for customers to interact with the restaurant, and a powerful administrative tool for the restaurant staff to manage operations efficiently. The application will serve as a digital storefront, offering features like online ordering, menu browsing, and customer feedback, while also providing a robust backend for managing meals, discounts, and feedback. This project aims to deliver a seamless digital experience for both customers and administrators, enhancing customer engagement and streamlining restaurant management.

2. System Requirements

- **User Interface:**
 - **Menu:** Users can view the full restaurant menu with categories, descriptions, and prices.
 - **Search:** A search functionality to find specific meals or categories.
 - **Order Customization:** Users can customize their orders (e.g., add extra toppings, specify dietary restrictions).
 - **Delivery System:** An integrated system for users to place delivery orders, providing address and contact information.
 - **Discounts:** The system will support special client discounts and student discounts, applied at checkout.
 - **Customer Feedback:** Users can submit feedback and ratings for their orders and experience.
- **Admin Interface:**
 - **Dashboard:** A central dashboard displaying key metrics like total orders, revenue, and recent feedback.
 - **Meal Management:** Admins can add, edit, and delete menu items, including images, descriptions, and prices.
 - **Feedback Tab:** A dedicated section to view and manage customer feedback.
 - **User Management:** Admins can manage user accounts and their discount eligibility.

3. System Design and Architecture

The application will be built using a client-server architecture.

- **Frontend (Client-Side):** This will be the user and admin interface, developed as a Single Page Application (SPA) for a smooth user experience.
- **Backend (Server-Side):** This will handle the business logic, database interactions, and API endpoints.
- **Database:** A relational or NoSQL database will be used to store data for users, orders, menu items, and feedback.
- **API:** A RESTful API will connect the frontend and backend, allowing them to communicate.

4. Features

User-Facing Features:

- **Menu Browsing:** Interactive and visually appealing menu.
- **Search Functionality:** Real-time search results.
- **Order Customization:** Options for modifying meal components.
- **Shopping Cart:** A persistent cart to manage orders before checkout.
- **Secure Checkout:** Integration with a payment gateway.
- **Discount Application:** Automatic application of eligible discounts.
- **Feedback Form:** A simple form for submitting reviews and ratings.

Admin-Facing Features:

- **Dashboard Analytics:** Visual charts and graphs for key metrics.
- **CRUD for Meals:** Full control over the restaurant's menu offerings.
- **Feedback Management:** A system to review, respond to, or archive feedback.
- **Order Management:** A view to track incoming and completed orders.

5. Technologies that will be used

- **Frontend (GUI Design):** JavaFX
- **Backend:** Java, utilizing Java Database Connectivity (JDBC) for database communication.
- **Database:** A SQL-based relational database (e.g., MySQL, PostgreSQL).
- **Server:** Apache Tomcat

6. Project Timeline

Phase | Tasks

1. **Planning & Design** | Requirement gathering, UI/UX design, database schema design. |

Estimated time : 1 Week

2. **Frontend Development** | Building the user and admin interfaces, connecting to the API. |

Estimated time : 3 Weeks

3. **Backend Development** | Setting up the server, creating the API, implementing authentication. | Estimated time : 2 Weeks

4. **Integration & Testing** | Integrating all components, comprehensive testing, bug fixing. |

Estimated time : 1 Week |

5. **Deployment & Review** | Deploying the application, final review, and documentation. |

Estimated time : 1 Week |

7. Conclusion

This project will result in a fully functional and modern web application that will significantly improve the customer experience and streamline the restaurant's operations. By focusing on a clean design, robust features, and a reliable architecture, the application will provide a valuable digital tool for both the restaurant and its patrons. The successful completion of this project will demonstrate proficiency in full-stack web development and project management.

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